(†) The rings should be regarded as commutative rings with identity.

Proposition 0.1. *: Suppose S is a muliplicative set for the ring A, we have the natural homomorphism $f : A \to S^{-1}A, a \mapsto \frac{a}{1}$. If I is an ideal of A, then the extended ideal of I about f is*

$$I^e = S^{-1}I = \{\frac{a}{s} | \, a \in I, s \in S\}.$$

Hence $I^e = S^{-1}A \Leftrightarrow I \cap S \neq \phi$.

Proof: By the definition, we know that

$$I^e = \{\sum_{i=1}^{n} f(a_i)\frac{x_i}{s_i} = \sum_{i=1}^{n} \frac{a_i x_i}{s_i} | \, a_i \in I, \frac{x_i}{s_i} \in S^{-1}A, n \in \mathbb{N}\}$$

Let $s = \prod_{i=1}^{n} s_i$ and $t_i = \prod_{j \neq i} s_j$, then we get

$$\sum_{i=1}^{n} \frac{a_i x_i}{s_i} = \frac{\sum_{i=1}^{n} a_i x_i t_i}{s} = \frac{a}{s}$$

where $a \triangleq \sum_{i=1}^{n} a_i x_i t_i \in I$. $\Box$

Proposition 0.2. *: Let $\mu : R \to R_1$ be a ring homomorphism, $\mathfrak{p}$ is a prime ideal of R. Then $\mathfrak{p}$ is a contracted ideal of a prime ideal in $R_1 \Leftrightarrow \mathfrak{p}^{ec} = \mathfrak{p}$.*

Proof: ($\Rightarrow$): Since $\exists \mathfrak{q} \in SpecR_1$ such that $\mathfrak{q}^c = \mathfrak{p}$, then $\mathfrak{p}^{ec} = \mathfrak{q}^{cec} = \mathfrak{q}^c = \mathfrak{p}$;

($\Leftarrow$): Let $S := \mu(R \backslash \mathfrak{p})$, then $1_{R_1} = \mu(1_R) \in S$ and for $\forall \mu(a), \mu(b) \in S = \mu(R \backslash \mathfrak{p})$ (i.e., $\forall a, b \in R \backslash \mathfrak{p}$), we know that $\mu(a)\mu(b) = \mu(ab) \in S$ since $ab \in R \setminus \mathfrak{p}$, which shows that S is a multiplicative set of R_1. $\mathfrak{p}^e$ is the extended ideal of $\mathfrak{p}$ in R_1. For $\forall x \in \mathfrak{p}^e$, since $\mathfrak{p}^{ec} = \mathfrak{p}$, thus $\exists y \in \mathfrak{p}^{ec} = \mathfrak{p}$ such that $\mu(y) = x$. Since $y \notin R \setminus \mathfrak{p}$, so $x = \mu(y) \notin S$, which means that $\mathfrak{p}^e \cap S = \phi$. So we know that the extended ideal of $\mathfrak{p}^e$ in $S^{-1}R_1$ is a proper ideal of $S^{-1}R_1$, thus there exists a maximal ideal $\mathfrak{m}$ of $S^{-1}R_1$ such that $\mathfrak{m}$ contains the extended ideal of $\mathfrak{p}^e$ in $S^{-1}R_1$. Denote the contracted ideal of $\mathfrak{m}$ in R_1 by $\mathfrak{q}$, so we know $\mathfrak{q} \in SpecR_1$ and $\mathfrak{q} \supseteq \mathfrak{p}^e, \mathfrak{q} \cap S = \phi$. $\mathfrak{q}^c = \mathfrak{p}$. $\Box$

Proposition 0.3. *: Let $A \subseteq B$ be rings, B is integral over A. Let $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$ be two prime ideals of B. If $\mathfrak{p}_1 \cap A = \mathfrak{p}_2 \cap A$ (i.e., $\mathfrak{p}_1^c = \mathfrak{p}_2^c$), then $\mathfrak{p}_1 = \mathfrak{p}_2$.*

Proof: $\mathfrak{q} = \mathfrak{p}_1 \cap A = \mathfrak{p}_2 \cap A$, let $S = A \setminus \mathfrak{q}$, consider the commutative diagram

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ S^{-1}A \arrow[r] & S^{-1}B \end{tikzcd}$$

then $S^{-1}\mathfrak{p}_1 \subseteq S^{-1}B$ and $S^{-1}\mathfrak{p}_1 \cap A = S^{-1}\mathfrak{q} = \mathfrak{q}A_{\mathfrak{q}}$. Recall that $\mathfrak{q}A_{\mathfrak{q}}$ is the unique maximal ideal of $A_{\mathfrak{q}}$, hence $S^{-1}\mathfrak{p}_1$ is a maximal ideal in $S^{-1}B$. Since $S^{-1}\mathfrak{p}_1 \subseteq S^{-1}\mathfrak{p}_2$, so we know that $S^{-1}\mathfrak{p}_1 = S^{-1}\mathfrak{p}_2$. Thus $S^{-1}(\mathfrak{p}_2/\mathfrak{p}_1) \cong S^{-1}\mathfrak{p}_2/S^{-1}\mathfrak{p}_1 = 0$. So if $\mathfrak{p}_2/\mathfrak{p}_1 \neq 0$, then $\exists x \in \mathfrak{p}_1 \cap S$, which is a contradiction. Thus we must have $\mathfrak{p}_2/\mathfrak{p}_1 = 0$, hence $\mathfrak{p}_1 = \mathfrak{p}_2$. $\square$

Proposition 0.4. *: Let $A \subseteq B$ be rings, B is integral over A. If $\mathfrak{p} \in SpecA$, then $\exists \mathfrak{q} \in SpecB$ such that $\mathfrak{q}^c = \mathfrak{q} \cap A = \mathfrak{p}$.*

Proof: Consider the commutative diagram

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\beta"'] & B \arrow[d, "\alpha"] \\ A_{\mathfrak{p}} \arrow[r, "f_{\mathfrak{p}}"] & B_{\mathfrak{p}} \end{tikzcd}$$

So we know that $B_{\mathfrak{p}}$ is integral over $A_{\mathfrak{p}}$. Since $p \in SpecA$, hence $A_{\mathfrak{p}}$ is a local ring. Let $\mathfrak{n}$ be a maximal ideal of $B_{\mathfrak{p}}$, then $\mathfrak{n} \cap A_{\mathfrak{p}}$ is a maximal ideal of $A_{\mathfrak{p}}$ and therefore $\mathfrak{n} \cap A_{\mathfrak{p}} = \mathfrak{p}A_{\mathfrak{p}}$. Denote $\alpha^{-1}(\mathfrak{n}) \triangleq \mathfrak{q}$, then by the commutativity of the diagram. we have $f^{-1}(\alpha^{-1}(\mathfrak{n})) = \beta^{-1}(f_{\mathfrak{p}}^{-1}(\mathfrak{n}))$. Since $f_{\mathfrak{p}}^{-1}(\mathfrak{n}) = \mathfrak{n}^c = \mathfrak{p}A_{\mathfrak{p}}$, thus $f^{-1}(\mathfrak{q}) = \beta^{-1}(\mathfrak{p}A_{\mathfrak{p}}) = \mathfrak{p}$, which means that $f^{-1}(\mathfrak{q}) = \mathfrak{q} \cap A = \mathfrak{p}$ and $\mathfrak{q} \in SpecB$. $\square$

Going Up

Theorem 0.1. *: Let $A \subseteq B$ be rings and an integral extension. Let $\mathfrak{p}_1 \subseteq \mathfrak{p}_2 \subseteq \cdots \subseteq \mathfrak{p}_n$ be a chain of prime ideals of A and $\mathfrak{q}_1 \subseteq \mathfrak{q}_2 \subseteq \cdots \subseteq \mathfrak{q}_m$ $(m<n)$ is a chain of prime ideals in B such that $\mathfrak{q}_i \subseteq A = \mathfrak{p}_i$ $(1 \leq i \leq m)$. Then $\exists \mathfrak{q}_{m+1}, \mathfrak{q}_{m+2}, \ldots, \mathfrak{q}_n \in SpecB$ such that $\mathfrak{q}_1 \subseteq \mathfrak{q}_2 \subseteq \cdots \subseteq \mathfrak{q}_n$ and $\mathfrak{q}_i^c = \mathfrak{q}_i \cap A = \mathfrak{p}_i$ $(1 \leq i \leq n)$.*

Proof: By induction. We prove the case that $(n, m) = (2, 1)$:

Since $\mathfrak{p}_1 \subseteq \mathfrak{p}_2$ and $\mathfrak{p}_1 \subseteq \mathfrak{q}_1, \mathfrak{q}_1 \cap A = \mathfrak{p}_1$, we know that $A/\mathfrak{p}_1 \hookrightarrow B/\mathfrak{q}_1$ is an integral extension. Since $\mathfrak{p}_2/\mathfrak{p}_1 \in Spec(A/\mathfrak{p}_1)$, hence $\exists \mathfrak{q}_2/\mathfrak{q}_1 \in Spec(B/\mathfrak{q}_1)$ such that $\overline{f}^{-1}(\mathfrak{q}_2/\mathfrak{q}_1) = \mathfrak{p}_2/\mathfrak{p}_1$.

$$\begin{tikzcd} A \arrow[r, "f"] \arrow[d, "\psi"'] & B \arrow[d, "\varphi"] \\ A/\mathfrak{p}_1 \arrow[r, "\overline{f}"] & B/\mathfrak{q}_1 \end{tikzcd}$$

thus $f^{-1}(\varphi^{-1}(\mathfrak{q}_2/\mathfrak{q}_1)) = f^{-1}(\mathfrak{q}_2) = \mathfrak{q}_2 \cap A$, on the other hand, $\psi^{-1}(\overline{f}^{-1}(\mathfrak{q}_2/\mathfrak{q}_1)) = \psi^{-1}(\mathfrak{p}_2/\mathfrak{p}_1) = \mathfrak{p}_2$. Thus we know that $\mathfrak{q}_2 \cap A = \mathfrak{p}_2$ and $\mathfrak{q}_1 \subseteq \mathfrak{q}_2$, where $\mathfrak{q}_2 \in SpecB$. $\square$

Proposition 0.5. *: Let $A \subseteq B$ be rings and C is the integral closure of A in B. Then for the multiplicative set S in A, $S^{-1}C$ is the integral closure of $S^{-1}A$ in $S^{-1}B$.*

Proof: For $\forall x \in C$, $\exists a_i \in A$ such that

$$x^n + a_1x^{n-1} + \cdots + a_n = 0$$

thus we get

$$(\frac{x}{s})^n + \frac{a_1}{s}(\frac{x}{s})^{n-1} + \cdots + \frac{a_n}{s^n} = 0 \ (\forall s \in S)$$

where $\frac{a_i}{s^i} \in S^{-1}A$. Hence $\frac{x}{s}$ is integral over $S^{-1}A$, i.e., $S^{-1}C \subseteq intcl_{S^{-1}B}(S^{-1}A)$;

Conversely, for $\forall \frac{b}{s} \in intcl_{S^{-1}B}(S^{-1}A)$, then $\exists \frac{a_i}{s_i} \in S^{-1}A$ such that

$$(\frac{b}{s})^n + \frac{a_1}{s_1}(\frac{b}{s})^{n-1} + \cdots + \frac{a_n}{s_n} = 0$$

let $t = \prod_{i=1}^n s_i \in S$, then $(ts)^n((\frac{b}{s})^n + \frac{a_1}{s_1}(\frac{b}{s})^{n-1} + \cdots + \frac{a_n}{s_n}) = 0$, i.e.,

$$(tb)^n + a_1'(tb)^{n-1} + \cdots + a_n' = 0$$

where $a_i' \in A$. Thus $tb \in C$ and therefore $\frac{b}{s}\frac{tb}{ts} \in S^{-1}C$, which shows that $intcl_{S^{-1}B}(S^{-1}A) \subseteq S^{-1}C$.

Thus we know that $S^{-1}C = intcl_{S^{-1}B}(S^{-1}A)$. $\square$

Remark 0.1. *: Here, the notation $intcl_B(A) :=$ the integral closure of A in B.*

Remark 0.2. *: By this propsotion, we know that integral closedness is a local property.*

Theorem 0.2. *: Let A be an integral domain, then TFAE:*
(1) A is integrally closed;
(2) $A_\mathfrak{p}$ is integrally closed for $\forall \mathfrak{p} \in SpecA$;
(3) $A_\mathfrak{m}$ is integrally closed for $\forall \mathfrak{m} \in MaxA$.

Proof: Let C be the integral closure of A in its fraction field $frac(A)$. Hence

A is integrally closed $\Leftrightarrow A = C \Leftrightarrow A \hookrightarrow C$ is surjective $\Leftrightarrow A_\mathfrak{p} \hookrightarrow C_\mathfrak{p}$ is surjective for $\forall \mathfrak{p} \in SpecA$.

By the proposition above, we know that $C_\mathfrak{p}$ is the integral closure of $A_\mathfrak{p}$. And we also know that

$A_\mathfrak{p} \hookrightarrow C_\mathfrak{p}$ is surjective $\Leftrightarrow A_\mathfrak{m} \hookrightarrow C_\mathfrak{m}$ is surjective for $\forall \mathfrak{m} \in MaxA$. $\square$

Let $A \subseteq B$ be rings and $I \subseteq A$ is an ideal of A. For $x \in B$, if $\exists n \in \mathbb{N}$ and $a_i \in I$ such that

$$x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0$$

we say that x is integral over I.

Proposition 0.6. *: Let $A \subseteq B$ be rings, I is an ideal of A, $C = intcl_B(A)$, $I^e = IC$ is the extended ideal of I in C. Then $intcl_B(I) = \sqrt{I^e}$.(Here, $\sqrt{J}$ is an alternative notation for the radical ideal $r(J)$.)*

Proof: Let $x \in C = intcl_B(A)$ be integral over I, i.e., $\exists a_i \in I$ such that

$$x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0$$

thus $x^n = -(a_1x^{n-1}+\cdots+a_{n-1}x+a_n) \in I^e = IC$, the extended ideal of I in C. Hence $x \in \sqrt{I^e} = \sqrt{IC}$;

Let $\forall x \in \sqrt{I^e} = \sqrt{IC}$, so $\exists n \in \mathbb{N}$ such that $x^n \in I^e = IC$, therefore $\exists b_1, b_2, \ldots, b_m \in C$ and $x_1, x_2, \ldots, x_m \in I$ such that

$$x^n = \sum_{i=1}^{m} b_i x_i$$

Consider $A[b_1, b_2, \ldots, b_m]$, thus $A[b_1, b_2, \ldots, b_m]$ is a finitely generated A-module sicne b_i are all integral over B. Since $x^n M \subseteq IM$, consider the map

$$\mu_{x^n} : M \to M$$

$$m \mapsto x^n m \ (\forall m \in M)$$

then $mu_{x^n}(M) \subseteq IM$. By Caylay-Hamilton Theorem, $\exists a_1, a_2 \ldots, a_r \in I$ such that

$$\mu_{x^n}^r + a_1\mu_{x^n}^{r-1} + \cdots + a_{r-1}\mu_{x^n} + a_r = 0$$

which implies that

$$x^{nr} + a_1 x^{n(r-1)} + \cdots + a_r = 0$$

thus $x \in intcl_B(I)$. $\square$

Proposition 0.7. : *Let $A \subseteq B$ be integral domains, A is integrally closed and B is integral over A. Let $x \in B$ be integral over an ideal I of A. Then $x \in B$ is algebraic over the fraction field $K = frac(A)$, furthermore if the minimal polynomial of x over K is given by*

$$f(t) = t^n + a_1t^{n-1} + \cdots + a_{n-1}t + a_n$$

then we have $a_i \in \sqrt{I}$.

Proof: It's clear to see that x is algebraic over $K = frac(A)$;

Let $x_1, x_2 \ldots, x_n$ be the roots of $f(t)$ (i.e., $f(x_i) = 0$). Let L be an extension field of K containing $x_1, x_2, \ldots, x_n$. Then we know that $f(t) = \prod_{i=1}^{n}(t - x_i)$. Moreover, we can know that x_i are all integral over I and the cofficients a_i of $f(t)$ are symmetric polynomials of $x_1, x_2, \ldots, x_n$. Thus a_i are all integral over I, hence a_i are all integral over A. Since A is integrally closed, so $a_i \in A$. By Proposition 0.6, we get $a_i \in intcl_B(I) = \sqrt{I^e} = \sqrt{I}$. $\square$

Going Down

Theorem 0.3. : *Let $A \subseteq B$ be integral doamins, A is integrally closed and B is integral over A. Let $\mathfrak{p}_1 \supseteq \mathfrak{p}_2 \supseteq \cdots \supseteq \mathfrak{p}_n$ be a chain of prime ideals of A, $\mathfrak{q}_1 \supseteq \mathfrak{q}_2 \supseteq \cdots \supseteq \mathfrak{q}_m$ ($m<n$) is a chain of prime ideals of B such that $\mathfrak{q}_i^c = \mathfrak{q}_i \cap A = \mathfrak{p}_i$ ($1 \leq i \leq m$). Then $\exists \mathfrak{q}_{m+1}, \mathfrak{q}_{m+2} \ldots, \mathfrak{q}_n \in SpecB$ such that $\mathfrak{q}_1 \supseteq \mathfrak{q}_2 \supseteq \cdots \supseteq \mathfrak{q}_n$ and $\mathfrak{q}_i^c = \mathfrak{q}_i \cap A = \mathfrak{p}_i$ ($1 \leq i \leq n$).*

Proof: By induction. We prove the case that $(n, m) = (2, 1)$:

Let $S := B \setminus q_1$, since B is an integral doamin, we know that the map

$$B \to S^{-1}B = B_{\mathfrak{q}_1}$$

$$b \mapsto \frac{b}{1}$$

is an injective homomorphism. Thus we can say that $A \subseteq B \subseteq B_{\mathfrak{q}_1}$. The extended ideal of $\mathfrak{p}_2$ in B is

$$\mathfrak{p}_2^e = B\mathfrak{p}_2 = \{\sum_{i=1}^{n} b_i x_i |\, b_i \in B, x_i \in \mathfrak{p}_2 \subseteq \mathfrak{p}_1 = \mathfrak{q}_1 \cap A\}$$

thus we know that $B\mathfrak{p}_2 \subseteq \mathfrak{q}_1$. So $B\mathfrak{p}_2 \cap S = \phi$ by the definition of S. Hence the extended ideal of $B\mathfrak{p}_2$ in $B_{\mathfrak{q}_1}$ is $S^{-1}(B\mathfrak{p}_2) = B_{\mathfrak{q}_1}\mathfrak{p}_2$, which is the extended ideal of $\mathfrak{p}_2$ in $B_{\mathfrak{q}_1}$ as well. By Proposition 0.2, we just need to prove that the contraction of $B_{\mathfrak{q}_1}\mathfrak{p}_2$ in A is $\mathfrak{p}$, i.e., $B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A = \mathfrak{p}_2$. Once we proved this, by the one-to-one correspondence $\mathfrak{q} \leftrightarrow S^{-1}\mathfrak{q}$ of prime ideals, where $\mathfrak{q} \cap S = \phi$, then $\exists \mathfrak{q}_2 \in SpecB$ such that $\mathfrak{q}_2 \cap S = \phi$ and the contraction of $S^{-1}\mathfrak{q}_2$ in A is $\mathfrak{p}_2$ and therefore $\mathfrak{q}_2 \subseteq \mathfrak{q}_1, \mathfrak{q}_2 \cap A = \mathfrak{p}_2$.

Now we prove that $B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A = \mathfrak{p}_2$:

For $\forall x \in B_{\mathfrak{q}_1}\mathfrak{p}_2 = S^{-1}(B\mathfrak{p}_2)$, then $\exists y \in B\mathfrak{p}_2$ and $s \in B \setminus \mathfrak{q}_1$ such that $x = \frac{y}{s}$. Since $\mathfrak{p}_2^e = B\mathfrak{p}_2$ is the extended ideal of $\mathfrak{p}_2$ in B. By Proposition 0.6, we know that

$$intcl_B(\mathfrak{p}_2) = \sqrt{B\mathfrak{p}_2} \supseteq B\mathfrak{p}_2$$

thus y is integral over $\mathfrak{p}_2$. By proposition 0.7, the mimimal polynomial of y in $K = frac(A)$ gives rise to the equation that

$$y^r + u_1 y^{r-1} + \cdots + u_r = 0 \ \ (*)$$

where $u_i \in \sqrt{\mathfrak{p}_2} = \mathfrak{p}_2$. For $x \in B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A$, then $s = yx^{-1}$ and $x^{-1} \in K$. Dividing $(*)$ by x^r, we get the minimal polynomial of s over K

$$s^r + v_1 s^{r-1} + \cdots + v_r = 0$$

where $v_i = \frac{u_i}{x^i}$, thus $x^i v_i = u_i \in \mathfrak{p}_2$. Since s is integral over A, by Proposition 0.7 we know that $v_i \in \sqrt{A} = A$. If $x \notin \mathfrak{p}_2$, then we must have $v_i \in \mathfrak{p}_2$ and therefore $s^r = -(v_1 s^{r-1} + \cdots + v_{r-1}s + v_r) \in B\mathfrak{p}_2 \subseteq B\mathfrak{p}_1 \subseteq \mathfrak{q}_1$, which implies that $s \in \mathfrak{q}_1$ (a contradiction!). So we have $x \in \mathfrak{p}_2$, hence $B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A \subseteq \mathfrak{p}_2$. It's clear that $B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A \supseteq \mathfrak{p}_2$, thus $B_{\mathfrak{q}_1}\mathfrak{p}_2 \cap A = \mathfrak{p}_2$. $\square$